Constellation

10 CFR 50.73

CCN: 23-78

November 7, 2023

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Peach Bottom Atomic Power Station (PBAPS) Unit 3
Subsequent Renewed Facility Operating License No. DPR-56
NRC Docket No. 50-278

Subject: Licensee Event Report (LER) 2023-001-00 Standby Liquid Control Pump Inoperable for Greater than LCO Window due to Gas Intrusion

The subject report is being submitted in accordance with 50.73(a)(2)(i)(B) for any operation or condition which was prohibited by the plant's Technical Specifications.

There are no commitments contained in this letter. If you have any questions, please contact the Peach Bottom Regulatory Assurance Manager, Mr. Wade Scott at (717) 456-3047.

Respectfully,

Jeremy C. Searer
Acting Plant Manager
Peach Bottom Atomic Power Station

Enclosure

cc: USNRC, Administrator, Region I
USNRC, Senior Resident Inspector
W. DeHaas, Commonwealth of Pennsylvania
S. Seaman, State of Maryland
B. Watkins, PSE&G, Financial Controls and Co-Owner Affairs

NRC FORM 366 (10-01-2023)

U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 03/31/2024

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

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4. Title

Standby Liquid Control Pump Inoperable for Greater than LCO Window due to Gas Intrusion

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
09	09	2023	2023	- 001 -	00	11	07	2023	NA	☐ 050
									NA	☐ 052

9. Operating Mode	10. Power Level
1	91

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20		10 CFR Part 50			
☐ 20.2201(b)	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(d)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2203(a)(1)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(2)(i)	☐ 20.2203(a)(4)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(ii)	**10 CFR Part 21**	☐ 50.69(g)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(iii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
			☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Wade Scott, Regulatory Assurance Manager	(717) 456-3047

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	BR	ACC	G250	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On September 9, 2023, during performance of a functional test, the Unit 3 ‘B’ Standby Liquid Control Pump did not produce discharge pressure. The pump was declared inoperable and Technical Specification Limiting Condition for Operation (LCO) 3.1.7 Condition B was entered. Troubleshooting identified that the pump and part of the discharge piping was not sufficiently filled with water. The most likely source of gas intrusion was the nitrogen filled accumulator located on the discharge piping between the pump and the discharge check valve. The piping was refilled and the accumulator was replaced, after which the pump was run successfully. The system was restored to operable and the LCO was exited on September 12, 2023. The gas leakage rate was slow and therefore, the system was likely inoperable longer than the LCO-allowed outage time (i.e., 7 days). Therefore, the condition is being reported under 10CFR 50.73(a)(2)(i)(B). There were no other systems which were inoperable during this identified condition that contributed to this event. There were no actual safety consequences as a result of this event.

NARRATIVE

Event Description

On September 9, 2023, during performance of a functional test, the Unit 3 'B' Standby Liquid Control (SLC, EIIS: BR) Pump (EIIS: P) did not produce discharge pressure. The pump was declared inoperable and Technical Specifications (TS) Limiting Condition for Operation (LCO) 3.1.7 Condition B was entered at 20:32 hours. Troubleshooting identified that the pump and part of the discharge piping was not sufficiently filled with water. The most likely source of gas intrusion was a leaking bladder inside the pneumatic-hydraulic accumulator (EIIS: ACC). The device is installed on the piping near the SLC pump discharge relief valve to dampen pulsations from the pumps to protect the system piping. Nominal accumulator pressure is between 325 psig and 450 psig. The accumulator bladder was replaced, after which the pump and discharge piping were filled and the pump was run successfully. The system was restored to operable and the LCO was exited on September 12, 2023 at 17:39 hours.

At the time of discovery, Unit 3 was operating in MODE 1 at approximately 91% power, due to end-of-cycle coast down.

Peach Bottom TS LCO 3.1.7 Required Action B.1 requires that a SLC subsystem be restored to operable status within 7 days or in accordance with the Risk Informed Completion Time Program. If this action is not met, the plant is required to be in MODE 3 within 12 hours and MODE 4 within 36 hours.

The most likely cause of the condition is gas leakage through the accumulator bladder. The estimated leak rate was small, because no visible sources of leakage were found on the accumulator bladder and the rate of pressure decay within the accumulator was slow. Failure analysis is being performed on the bladder, but at this time, the exact cause of the leakage is unknown. The 3B SLC subsystem was last run successfully on May 29, 2023. Trending of available accumulator pressure data did not provide a conclusive time of failure. Due to the slow nature of the leakage and the volume of gas intrusion present, it is highly likely that the condition existed longer than the LCO-allowed outage time.

Since the 3B train of SLC was likely inoperable for greater than the LCO-allowed outage time, this condition is being reported under 10CFR 50.73(a)(2)(i)(B).

The Energy Industry Identification System component function identifier and name of each component or system:
EIIS: Standby Liquid Control System, BR
3BT076, Standby Liquid Control N2 Accumulator
Manufacturer: Greer Hydraulics, Inc
Model: A68514-200

Safety Consequences

The SLC System is designed to provide the capability of bringing the reactor, at any time in a fuel cycle, from full power and minimum control rod inventory (which is at the peak of the xenon transient) to a subcritical condition with the reactor in the most reactive, xenon free state without taking credit for control rod movement. The SLC System satisfies the requirements of 10 CFR 50.62 on anticipated transient without scram using highly enriched boron. The SLC System is also used to maintain suppression pool pH at or above 7 following

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NARRATIVE

a loss of coolant accident (LOCA) involving significant fission product releases. The SLC system consists of two redundant trains, either of which is sufficient to perform its safety function.

There were no other systems which were inoperable during this identified condition. The 3A SLC train demonstrated full functional capability by performance of a surveillance test on September 3, 2023. There were no actual safety consequences as a result of this event, because the redundant train maintained operability.

Corrective Actions
The Unit 3 ‘B’ SLC accumulator bladder was replaced to restore the subsystem to full function. Investigation into the cause of the bladder leakage is ongoing. If the results of this investigation substantially change the assessment of this condition or planned corrective actions, a supplement to this LER will be submitted.

The remaining SLC discharge lines (2A, 2B, and 3A) were examined by ultrasonic testing on October 9, 2023. All three discharge lines contained gas voids, which were significantly less than what was discovered on the 3B line. None of the discovered gas voids extended to the pump casings or internals. The 3A pump was operated with gas voids present, successfully demonstrating that this volume of gas intrusion posed no adverse impact to the pump function. The 3B pump was also examined at this time, and the piping was full of water. The SLC pump discharge process piping will be periodically monitored until replacement accumulator bladders can be procured and installed.

Previous Similar Events
Peach Bottom has not previously experienced a failure of a SLC system accumulator bladder that adversely impacted pump performance.